

Summary of Changes

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General response

We would like to express our deep gratitude to the editor and the reviewers for their constructive comments on our paper titled **“FireKARE: Multimodal Biomechanical Risk Assessment for Incident Knee and Ankle Injuries in Firefighters”**, ID: IoT-67039-2026. We have revised our manuscript according to the feedback. Please find our responses below.

In this revised version, we accept all reviewers’ suggestions and comments to improve the paper, and fix the grammatical and sentence construction errors carefully to make the paper read more smoothly. All the questions, concerns, and comments have been addressed and incorporated into our revision and response letter. We have thoroughly considered the comments of each reviewer point by point. [For convenience, modifications are made and highlighted in the revised manuscript.](#) We hope the reviewers are satisfied with the changes we made and wish our paper to be reconsidered for further arrangements.

The followings are the comments of the reviewers and our replies.

1 Response to Associate Editor

Comment 1

The innovation should be further highlighted. Moreover, some concepts need further clarification.

Response:

2 Response to Reviewer #1

We would like to express sincere thanks to the reviewer for the thorough and constructive comments. In what follows, we present detailed comments in response to the individual points raised by the reviewer and elaborate on how the manuscript has been revised.

Comment 1

The literature review section is rather limited. The authors should provide a more comprehensive review of related work.

Response: Thank you for pointing this out.

Comment 2

Fig. 1 appears somewhat AI-generated and contains an excessive amount of information. It is recommended that the authors redraw the figure as a double-column illustration and highlight the key components more clearly.

Response: Thank you for pointing this out.

Comment 3

In Sec. II-A, the “domain-knowledge-guided strategy” is not clearly defined. The authors do not provide a detailed explanation of this strategy, nor cite relevant references to support it.

Response: Thank you for pointing this out.

Comment 4

The authors do not provide a sufficiently detailed description of the dataset. Were the data collected by the authors using multiple sensors, or were they obtained from existing public datasets? Clearly specifying the data source would better demonstrate the connection between the study and IoT, and would make the paper more aligned with the scope of the journal.

Response: Thank you for pointing this out.

3 Response to Reviewer #2

Response: We would like to express sincere thanks to the reviewer for the thorough and constructive comments. In what follows, we present detailed comments in response to the individual points raised by the reviewer and elaborate on how the manuscript has been revised.

Comment 1

The central claim of prospective injury-risk warning is not convincingly supported by a cohort of only 40 firefighters with 11 injury-positive cases.

Response: Thank you for your valuable suggestion.

Comment 2

The construction of the 20 mechanism-aware biomarkers from 658 descriptors is insufficiently specified.

Response: Thank you for your valuable suggestion.

Comment 3

The sensing setup relies on laboratory-style measurements without demonstrating deployability, robustness, or cost feasibility in real firefighter training environments for IoT applications.

Response: Thank you for your valuable suggestion.

4 Response to Reviewer #3

Response: We would like to express sincere thanks to the reviewer for the thorough and constructive comments. In what follows, we present detailed comments in response to the individual points raised by the reviewer and elaborate on how the manuscript has been revised.

Comment 1

I am afraid this paper is not enough to use "IoT framework". The claim of an IoT framework should be weakened. It is more appropriate to be defined as IoT sensing-enabled biomechanical risk assessment application.

Response: Thank you for your valuable suggestion.

Comment 2

It is indeed difficult to get many samples. It shows that accuracy improves by 7.5 percentage points over the best baseline, but the AUROC gain over RBF-SVM is only 0.017. With only 40 subjects, statistical significance is not established.

Response: Thank you for your valuable suggestion.

Comment 3

The authors must clarify whether the selection of the 20 biomarkers introduces information leakage. That is really important for the model.

Response: Thank you for your valuable suggestion.

Comment 4

I think ablation studies are necessary. The authors should evaluate the contribution of the critical component.

Response: Thank you for your valuable suggestion.

Comment 5

The related work section should include more recent wearable/IoT-based biomechanical injury prediction studies.

Response: Thank you for your valuable suggestion.

Comment 6

Table I should include 95% confidence intervals.

Response: Thank you for your valuable suggestion.

References